VITALINK ? Offline Maternal Assessment & Auditable Referral Handoff

Deterministic clinical safety rules · Offline-first referral outbox · Regional clinician command center

- Research prototype using synthetic maternal records (SYN-*) only; not an approved clinical device.
- Bridges community health worker (CHW) field screening and hospital clinician acknowledgment.
- 221/221 automated tests passing across Kotlin/Android, Java 21/Spring Boot, Next.js/Playwright, and Python.
- Debug APK (app-debug.apk), Spring Boot API JAR, and Next.js standalone production bundle built and verified.

DEMO READY WITH DOCUMENTED LIMITATIONS

A Recorded Blood-Pressure Reading Does Not Prove Clinician Receipt

Field screening fails when offline capture, server receipt, and clinician review are conflated.

- Intermittent rural connectivity interrupts referral transmission during urgent maternal escalations.
- Naive network retries risk duplicate referrals or silent drops without transactional outbox guarantees.
- Server HTTP receipt is not the same event as a receiving doctor reviewing and acknowledging the patient.
- VITALINK enforces strict separation: Saved Locally != Server Received != Clinician Acknowledged.

Measure -> Evaluate -> Escalate -> Queue -> Synchronize -> Acknowledge

Immediate local escalation guidance never waits for network connectivity.

- 1. Offline Maternal Assessment: Validated systolic/diastolic BP, gestational age, source metadata, and red-flag symptoms.
- 2. Deterministic Safety Evaluation: Immediate on-device triage with explicit trigger reasons.
- 3. Durable Local Outbox: AES-256-GCM encrypted payload persisted in Room with stable UUID + Idempotency-Key.
- 4. Regional Command Center Sync: Facility-scoped queue, clinician review, and auditable acknowledgment.

Systolic >= 160 OR Diastolic >= 110 mmHg -> Urgent Clinical Assessment

Versioned rule specification (vitalink-rules-1.0.0) verified across 4 languages.

- Severe BP uses strict OR semantics: verified at exact boundaries 159/109, 160/109, 159/110, and 160/110.
- Red-flag maternal symptoms (severe headache, visual disturbance, epigastric pain, etc.) remain URGENT with normal BP.
- Missing or out-of-range measurements trigger REVIEW_REQUIRED ? never suppressed as low-risk.
- 16/16 canonical shared test cases pass identically in Kotlin, Java 21, TypeScript, and Python.

Transactional Outbox, Keystore AES-GCM & Non-Regressing State Machine

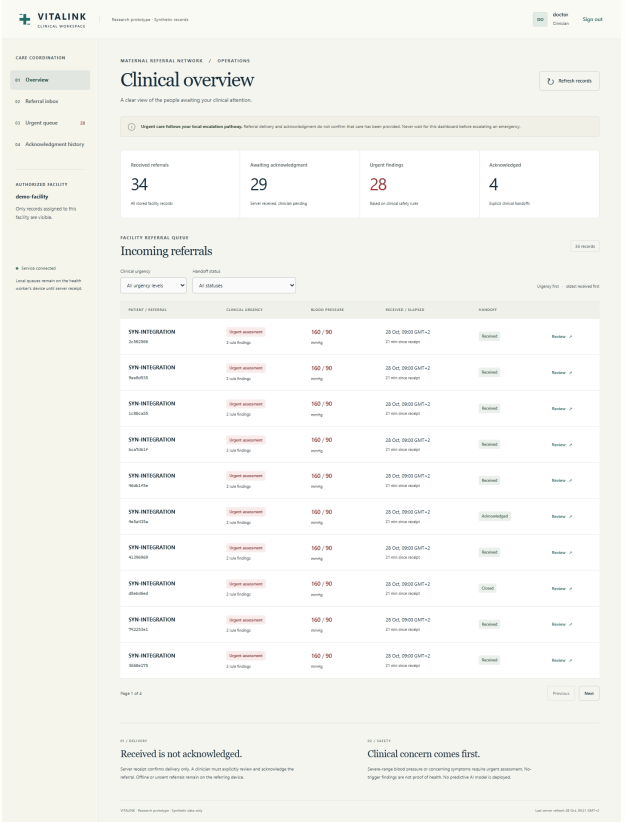
Android Kotlin + Jetpack Compose + Room + WorkManager (app-debug.apk built & verified)

- State lifecycle: DRAFT -> PENDING_SYNC -> SENT -> SERVER_RECEIVED -> ACKNOWLEDGED -> CLOSED.
- SENT represents an in-flight transmission attempt; hospital receipt is never shown before verified server receipt.
- Sensitive assessment/referral JSON encrypted via AndroidKeyStore AES/GCM/NoPadding (96-bit IV, 128-bit tag, AAD).
- 64/64 Android Kotlin unit tests pass; host-JVM rule latency median 1.0 us (p95 1.2 us).

Java 21 / Spring Boot / PostgreSQL 18.1 + Next.js 16 Doctor Workspace

Live browser captures from Playwright E2E suite (0 axe-core WCAG 2.1 AA violations)

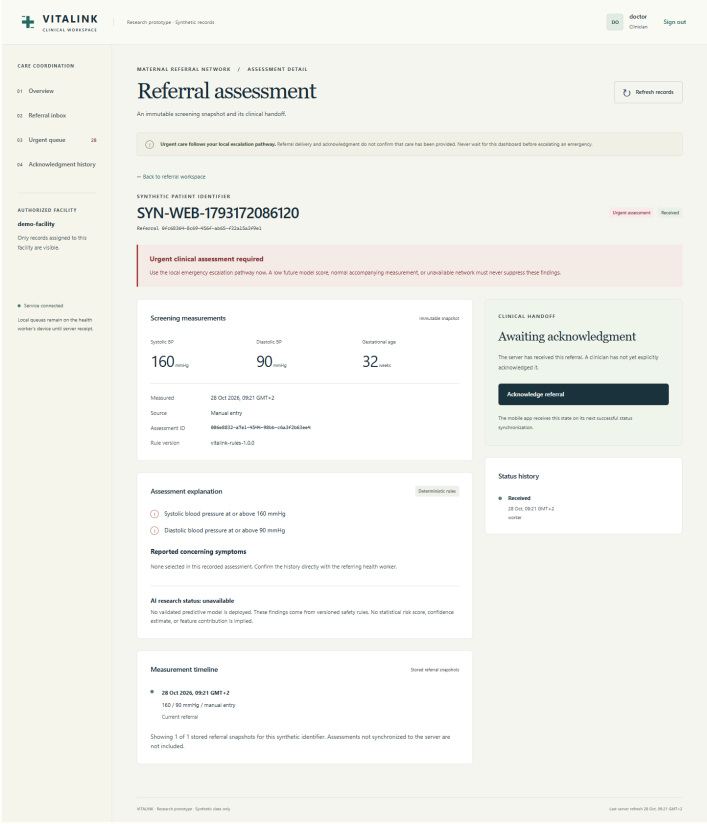
- 57/57 backend JUnit tests pass (including Testcontainers postgres:17-alpine & 25 PostgreSQL integration tests).
- 36/36 black-box HTTP lifecycle tests & 6/6 Playwright browser E2E tests pass against live API + DB.



Deterministic Trigger Explanations & Auditable Clinician Handoff

Verified across desktop (1440x960) and mobile (390x844) viewports with zero horizontal overflow.

- Displays exact BP readings (e.g., 160/90 mmHg), gestational weeks, source, and human-readable safety trigger reasons.
- Requires explicit two-step confirmation before recording ACKNOWLEDGED status with clinician actor & note.



Current Release: Deterministic Rules-Only with Safe Predictive Abstention

Refuses training on clinically inadequate public datasets (Exit Code 2: BLOCKED).

- Evaluated UCI Maternal Health Risk (ID 863): lacks clinical outcome definitions, gestational timing, and patient IDs.
- Automated provenance gate (ml/vitalink_research/pipeline.py) blocks model training and prevents synthetic label leakage.
- Runtime returns MODEL_UNAVAILABLE and abstains safely; no fabricated accuracy, ROC-AUC, or ONNX claims.
- 27/27 Python mathematical unit tests pass (patient-disjoint splits, Brier/ECE calibration, bootstrap CI).

Illustrative Unit Allocation: INR 7.78 per Screening (Scenario Only)

Transparent formula with explicit exclusions ? no unmeasured clinical outcome claims.

- Formula: INR 10,000 cuff kit / (24 months x 150 screenings/month) [= INR 2.78] + INR 5.00 consumables = INR 7.78.
- Volume sensitivity: INR 15.00 at 50/mo; INR 9.17 at 100/mo; INR 7.78 at 150/mo; INR 6.39 at 300/mo.
- Excludes CHW stipends, transport, hospital staffing, device replacement, training, and server hosting.
- Pilot evaluation metrics: outbox sync completion rate, receipt-to-acknowledgment latency, duplicate prevention.

221/221 Automated Tests Passing · Ready for Live Hackathon Demonstration

Complete transparency on verified capabilities and remaining environmental boundaries.

- Verified PASS: M1 Clinical Foundation (64 Kotlin + 26 Java + 31 TS + 27 Python tests), M3 Backend & Web (57 JUnit + 6 E2E).
- Verified PARTIAL / BLOCKED: M2 on-device ADB instrumentation & M4 predictive model training (no authorized clinical cohort).
- Reproducible 3-minute live demo: severe assessment -> idempotent sync -> live dashboard acknowledgment.
- Clinical deployment requires formal clinical governance, ethics approval, and prospective cohort validation.